

Functional Requirement

Order Management

- FR-ORD-01** Manager can create a new production order with:
customer name, cup quantity, cup type, expected delivery date
- FR-ORD-02** System auto-checks raw material availability on order creation
against required quantities (BOM lookup)
- FR-ORD-03** If stock is sufficient → order moves to RAW_MATERIAL_ISSUED
If stock is insufficient → order is flagged as PENDING_STOCK
- FR-ORD-04** Each order has a unique Order ID used for tracking
across all modules
- FR-ORD-05** Manager can view all orders with their current production stage
- FR-ORD-06** Order stage progression is driven by stage-specific forms
submitted by Floor Supervisor or Worker.
Each form captures:
- Order ID / Batch Name
 - Product made and quantity produced
 - Date (auto-filled by system)
 - Name of Supervisor submitting the form
 - Any stage-specific relevant details

Stage advances automatically when the quantity reported
in the form matches the order's required quantity.

Each production stage has its own dedicated form.

Inventory and Raw material Management

- FR-INV-01** Manager can stock-in raw materials with:
quantity, type, unit, supplier name, date of entry

- FR-INV-02** All stocked-in material enters as UNRESERVED

- FR-INV-03** When an order is confirmed, system reserves required
quantities against that Order ID

- FR-INV-04** Reserved stock is locked — cannot be assigned to another order

- FR-INV-05** Manager can view stock dashboard showing per material type:
total stock / reserved stock / available (unreserved) stock

- FR-INV-06** System flags a low-stock alert when available quantity
of any material falls below a defined minimum threshold

- FR-INV-07** Every stock movement is logged with:
movement type (IN / RESERVED / CONSUMED),
timestamp, actor who performed it, and linked Order ID

Production Flow Management

FR-PRD-01 Every production order progresses through exactly 9 stages

in strict forward order — no skipping, no reversal:

ORDER_RECEIVED -> RAW_MATERIAL_ISSUED -> SHEET_MAKING_IN_PROGRESS -> SHEET_READY -
> CUP_MOLDING_IN_PROGRESS -> CUPS_READY_FOR_PRINTING -> PRINTING_IN_PROGRESS ->
READY_TO_DISPATCH -> DISPATCHED

FR-PRD-02 Each stage is advanced by a dedicated form submitted by

Floor Supervisor or Worker.

Stage advances only when reported quantity = order quantity.

Only forward progression is allowed.

FR-PRD-03 Each stage update records:

submitted_by (employee), timestamp, quantity reported,
and optional notes

FR-PRD-04 Manager and Admin can view the full stage history

of any order as a complete audit trail

FR-PRD-05 Floor Supervisor can only see and update orders

that belong to their currently active shift

Dispatch Management

FR-DSP-01 Only orders in READY_TO_DISPATCH stage can be dispatched

FR-DSP-02 Only Admin or Manager can confirm dispatch —
via a single confirm action (button/click)

FR-DSP-03 On dispatch confirmation:

- Order status moves to DISPATCHED
- All raw material reservations for that order
are marked as CONSUMED

FR-DSP-04 Dispatch records are immutable once confirmed

FR-DSP-05 Manager and Admin can view dispatch history
filtered by date and order

Employee Management

FR-EMP-01 HR Manager can create, update, and deactivate employee records

FR-EMP-02 Employee master record contains:
name, role, department, date of joining,
base salary, contact details, active status

FR-EMP-03 Shift assignments are maintained in a separate shift
assignment table linked to the employee master via FK (employee_id)
This allows clean separation of identity data from
scheduling data and supports shift history tracking

FR-EMP-04 HR Manager can assign and reassign employees to shifts
via the shift assignment table

FR-EMP-05 Admin can view all employee records
Manager has no access to the employee module

Attendance Management

- FR-ATT-01** Attendance is recorded automatically via
clock-in / clock-out punches

- FR-ATT-02** System calculates daily hours worked per employee
based on clock-in and clock-out timestamps

- FR-ATT-03** System automatically marks daily attendance status:
Present / Absent / Late / Half-Day
based on shift start time and hours worked rules

- FR-ATT-04** Only Admin and HR Manager can manually override
an attendance record.
Every override is logged with: reason, actor, and timestamp

- FR-ATT-05** HR Manager can view attendance reports filtered by:
employee / date range / department

- FR-ATT-06** Attendance records feed into monthly payroll calculation

Payroll Management

- FR-PAY-01** HR Manager can generate monthly payroll
for all active employees
- FR-PAY-02** Payroll is calculated as:
base salary + (attendance days × daily rate) + overtime amount - deductions
- FR-PAY-03** Once marked as FINALIZED, a payroll record is immutable —
no edits or deletions allowed
- FR-PAY-04** HR Manager can view payroll history
per employee per month
- FR-PAY-05** Admin has read-only access to all payroll records

Authentication & Access Control

- FR-AUTH-01** All users must authenticate via email and password
- FR-AUTH-02** Role-based access control (RBAC) is enforced at the API level —
UI restrictions alone are not sufficient
- FR-AUTH-03** Sessions are managed via JWT tokens
- FR-AUTH-04** Admin can create user accounts and assign roles
- FR-AUTH-05** Passwords are stored as hashed values only —
plain text storage is strictly prohibited